

Argmax Decoding Causes Liquidity Failure in a Market of Language-Model Traders

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Abstract

We populate a continuous double auction with traders whose decisions are made by a typed language model that returns a probability distribution over actions, and vary only how that distribution is consumed. Taking the model’s modal action—the default in virtually every deployment—makes the market stop trading after shocks: following a positive shock, the market clears in 66.6% of periods against 93.8% after a negative shock, a gap of 25.8 percentage points (95% CI [17.3, 34.2], $t = 6.37$, $n = 20$ preregistered). Sampling from the same distribution reduces the gap to 3.1 points. Two algorithmic baselines that are symmetric by construction show no gap. The mechanism is direct and observable: 100% of non-trading periods under modal decoding had one side of the order book entirely empty, against 3.4% for the zero-intelligence baseline. Because every trader conditions on the same public signal, a deterministic policy makes them act identically, and a market of unanimous buyers has no counterparty.

We test this mechanism with a preregistered prediction it could have failed: if unanimity is the cause, the effect must vanish as the market grows. It does, from 37.0 points at four traders to 4.2 at thirty-two (-0.112 per doubling, $t = -6.91$). A companion prediction—that sampling would show no such decay—was falsified: sampling herds too, seven times more weakly, on the same curve. We therefore report a single mechanism whose strength is set by how sharply the decode rule collapses the model’s distribution, rather than two regimes.

Separately, we find that ordinary domain wording induces a 25-point asymmetry in the model’s stated conviction between buying and selling at equal mispricing, which vanishes under mirror-image phrasing. A preregistered test found this does *not* move pricing error, and we report that null.

All experiments were preregistered, including the two hypotheses that failed.

1 Introduction

A language model asked to choose among actions returns, in principle, a distribution. Almost every deployment discards it and takes the mode. For a single decision this is correct: the mode is the most likely action. This paper asks what happens when many agents driven by the same model make related decisions at the same time.

The answer, in a market, is that the market stops. Modal decoding makes agents conditioning on shared information act identically. In a setting that requires counterparties, unanimity is not consensus; it is the absence of trade. We measure this in a continuous double auction, isolate the mechanism, and confirm it with a prediction that could have failed and did not.

Our contributions:

1. **A liquidity failure caused by decoding, not by reasoning.** The model’s modal action is *correct* on both sides at every mispricing we tested. Nothing about its judgement is wrong. The failure is entirely in the aggregation of correct decisions.
2. **A directly observed mechanism.** We instrument both sides of the order book and show that under modal decoding, non-trading periods are one-sided essentially always, while for symmetric baselines they are not.
3. **A falsifiable size-scaling test.** The mechanism implies the effect must decay with market size at a rate set by the agreement rate. We registered this before running it; it held.
4. **A prompt-sensitivity result, and a null.** Domain-idiomatic wording induces a large conviction asymmetry invisible to anyone reading only the modal action. A preregistered test found it does not move pricing error.

We emphasise the failures. Three preregistered predictions in this project were wrong, and each is reported: the original mechanism (Section 5), the sampling-amplification prediction, and the claim that sampling would show no size decay (Section 7). The result that survived did so against tests designed to kill it.

2 Related work

[TO BE COMPLETED. This section requires a literature search the authors have not yet performed; citations must be verified before submission. The two references below are load-bearing for the design and are believed accurate, but should be checked against the published record.]

Our market design follows the experimental tradition of Smith [1962] and takes its lower baseline from the zero-intelligence traders of Gode and Sunder [1993], whose central result—that budget-constrained random traders achieve high allocative efficiency—motivates our use of a random arm as a floor rather than a strawman.

Work that must be surveyed before submission: LLM-populated economic simulations; herding and information cascades in financial markets; decoding strategies and their effect on output diversity; calibration of language-model confidence.

3 The instrument

3.1 Market

A continuous double auction in one abstract good. Prices are integer ticks; matching is price-then-time priority; an incoming order executes at the resting order’s price. A trader’s resting orders are withdrawn at the top of its turn, so each trader has at most one live order. Submitting on one side cancels the trader’s resting orders on the other, preventing wash trades without leaving a crossed book.

Accounts are conserved by construction. Each trade moves cash from buyer to seller and inventory from seller to buyer, one for one. Short sales and borrowing are disallowed, and every order is pre-checked against uncommitted cash or uncommitted inventory. A property test submits 20,000 random orders and asserts after every one that total cash and total units are exactly their endowed values, that no account is negative, that no trader has committed more than it holds, and that the book is never crossed.

Order size is fixed at one unit, so aggressiveness is the only lever an arm has; position sizing is a second channel and is deliberately disabled.

3.2 Fundamental and information

The fundamental F_t is piecewise constant. For the size and liquidity experiments we use a *matched-jump* process: jumps are evenly spaced, of identical magnitude, strictly alternating in direction, and even in number, so that every up window is paired with a down window of the same size between the same two price levels. This makes the up-versus-down comparison within-run and matched, which matters: under a random-jump process the per-run noise on this statistic has standard deviation 0.74, large enough that a single seed of the symmetric baseline returned a spurious -0.82 gap.

Each trader’s private value is $F_t + \varepsilon_{i,t}$ with $\varepsilon \sim \mathcal{N}(0, 5)$, redrawn each period. All experiments here use full information; the delayed and noisy signal treatments are implemented but not reported.

3.3 The decision contract

Code owns matching, budgets, order sizes and prices. An agent answers three typed questions and nothing else: direction (**buy/sell/pass**), urgency on a five-level scale, and whether the signal is already reflected in the book. A code-owned rule maps urgency to an integer tick, anchored on the book and hard-clamped at the agent’s own private value.

That clamp is load-bearing. *An agent cannot express a losing price*, because it never expresses a price. A 20,000-case property test asserts that no quote is ever on the wrong side of the quoting agent’s value. Consequently every effect we report is about direction and timing, never about arithmetic.

The question wording is frozen and hash-locked; a test fails if any instruction or criterion changes. This is the defence against unrecorded prompt iteration.

3.4 Arms

All four arms emit the same object, including a distribution over **{buy, sell, pass}**, which makes them comparable.

- **ZI**: random direction and urgency, fixed policy distribution, value-clamped so it never knowingly trades at a loss.
- **NBR**: logit best-response over the same urgency grid, scored by expected surplus times a fill probability. Exactly symmetric between sides by construction; a buy at mispricing $+x$ and a sell at $-x$ receive identical probability.
- **Jev-argmax**: the model’s modal action.
- **Jev-sample**: a draw from the model’s distribution.

The two model arms read one response, so for a given observation they are one API call and differ only in the decode rule.

4 The model and its measurement properties

The model is a commercially available typed-decision model that returns, for a multiple-choice question, a selected option together with a probability over all options. Three properties affect measurement and are reported because they constrain what can be claimed.

Non-determinism. Identical requests return different answers. Four repetitions of one request returned urgency scores of 2.84, 2.79, 2.82 and 2.74 and buy probabilities of 0.98, 0.99, 0.99, 0.99. There is no seed parameter. Run-to-run noise is roughly ± 0.05 on a 0–4 urgency scale and ± 0.01 on probabilities. *Reproducibility therefore requires the response archive, not merely the code:* a rerun without it does not reproduce. All 110,805 responses are retained and every reported figure is regenerable from them.

Quantisation. Probabilities are returned to two decimal places, so three-option distributions routinely sum to 0.99 or 1.01. We accept and renormalise within tolerance.

Undocumented failure modes. The published status codes are incomplete; the service returns bare 5xx from its edge under load. A client written to the documentation alone will treat these as fatal.

5 Experiment 1: wording, conviction, and a null

5.1 Domain wording induces a conviction asymmetry

Our first criteria were written in ordinary domain language: *buy* as “the good is underpriced... you want to acquire a unit”, *sell* as “the good is overpriced... you want to dispose of a unit”. At equal absolute mispricing the model placed 0.988 on the profitable side when buying was correct and 0.738 when selling was correct, a gap of 0.250, with the missing mass on *pass*.

The gap survives a second price level (+0.225), reversing the option order (+0.235), and renaming the options neutrally (+0.185). It does not survive rewriting the two criteria as exact structural mirrors—“value is *above* the price” versus “*below* the price”—under neutral instructions, where the gap is +0.005 (Figure 1).

Two points matter for practice. First, the modal action is *correct on both sides at every point tested*: a practitioner reading only the selected option sees nothing wrong. Second, the wording that induces the skew is the natural one; symmetry had to be engineered deliberately.

5.2 It does not move pricing error

We predicted this skew would slow price discovery after negative shocks. A four-seed exploratory run supported it (+0.63, four of four seeds positive, Holm $p = 0.061$). We registered the test and ran it on twenty fresh seeds. It failed: -0.184 , 95% CI $[-0.479, +0.111]$, seven of nineteen seeds positive, one-sided $p = 0.897$. The confirmatory interval excludes the exploratory estimate. With SE = 0.140 the run had ample power to detect the exploratory effect, so this is evidence of absence.

We report it as such. Wording skews what the model says; it does not, in this market, change what prices do.

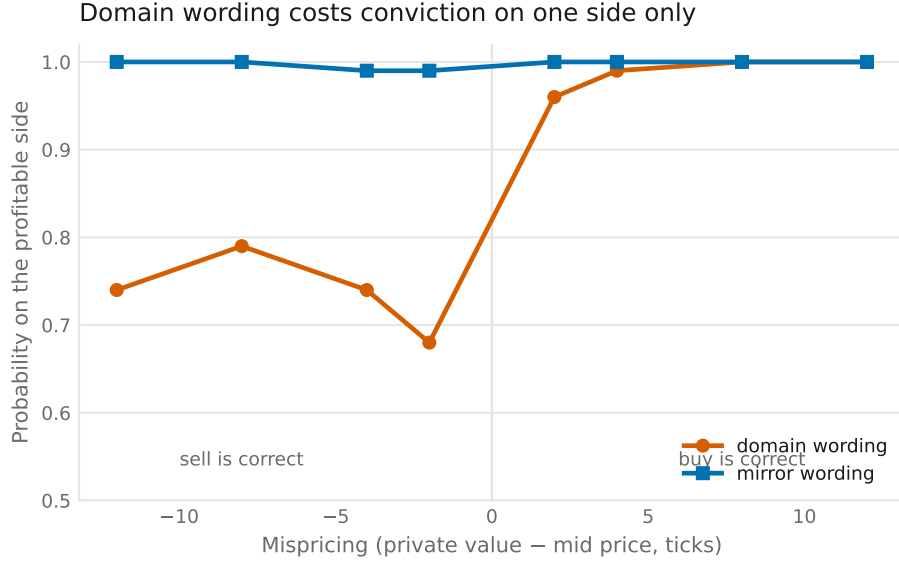


Figure 1: Probability the model places on the profitable side, against mispricing, under the two wordings. The model is symmetric when asked symmetrically. The effect is $25\times$ the model’s own run-to-run noise.

6 Experiment 2: liquidity failure under modal decoding

The null above was correct for the question it asked, and the question was wrong. Root mean squared pricing error is defined only over periods in which trade occurred, and so discards exactly the periods where the effect lives. In one run, modal decoding produced zero trades in all three up-jump windows and full trading in all three down-jump windows—270 trades in the run, none in the 45 periods after a positive shock.

We therefore registered a new outcome, defined for every period: $\text{traded}(d)$, the share of the fifteen periods after a jump of direction d in which at least one trade occurred.

Mechanism, observed. We instrument resting depth on both sides at the close of every period, which separates two failures: *no counterparty* (one side empty) and *no crossing* (both sides present, prices do not meet). Of the silent post-shock periods:

Arm	after up	after down
Jev-argmax	100.0%	95.6%
Jev-sample	71.9%	56.2%
ZI	3.4%	5.3%
NBR	9.1%	0.0%

Every silent period under modal decoding after a positive shock had no counterparty whatsoever. The symmetric baselines fall silent for the ordinary reason instead.

A control that does not quite hold. NBR returns +1.0 points ($t = 3.33$), small but distinguishable from zero, indicating a slight genuine structural tilt in the market. It is $1/26$ of

Arm	Traded share		Gap
	after up	after down	
Jev-argmax	66.6%	93.8%	+25.8
Jev-sample	93.0%	93.9%	+3.1
ZI	77.3%	75.0%	−0.4
NBR	99.8%	99.3%	+1.0

Table 1: Preregistered confirmatory test, twenty fresh seeds. D1: the Jev-argmax gap exceeds zero, $t = 6.37$, eighteen of twenty seeds positive. D2: it exceeds the Jev-sample gap by 22.7 points, $t = 6.01$. Both Holm-significant.

the effect under test, ZI is clean (-0.4 , $t = -0.22$), and it does not survive the per-size controls of Section 7. We report it because the registration required controls to sit on zero and this one does not.

7 Experiment 3: the mechanism, tested where it could fail

If the market halts because every trader reaches the same conclusion, the effect must depend on the number of traders. With agreement rate p measured independently at 0.92, the probability that all N agree is approximately p^N , which falls from 0.72 at $N = 4$ to 0.07 at $N = 32$. We registered two predictions before running: that the argmax gap would decay with N (E1), and that the sampling gap would not (E2).

E1 confirmed. The gap falls from 37.0 points at $N = 4$ to 4.2 at $N = 32$; slope -0.112 per doubling, $t = -6.91$. It is robust to dropping the $N = 4$ point, which we flagged in advance as floor-constrained because the symmetric baselines themselves trade in only 33% of periods there (-0.108 , $t = -4.21$).

E2 falsified. We predicted sampling would show no decay. It shows -0.016 per doubling, $t = -2.53$. The available rescue—attributing it to the floor-constrained $N = 4$ point—fails: dropping that point makes the decay *more* significant ($t = -3.53$).

The corrected account is better than the prediction. Sampling herds as well, seven times more weakly, on the same curve. At a buy probability of 0.99, most sampled traders still choose to buy; sampling does not abolish correlation but dilutes it. There is one mechanism, whose strength is set by how sharply the decode rule collapses the distribution, rather than two regimes.

8 Discussion

The practical claim is narrow and, we think, general in shape: *the best way to use one model is not the best way to use many of them*. Modal decoding is individually optimal and collectively degenerate whenever agents share information and require counterparties or diversity. Sampling is individually worse—it deliberately takes a less likely action some of the time—and here it was the difference between a market that cleared and one that did not.

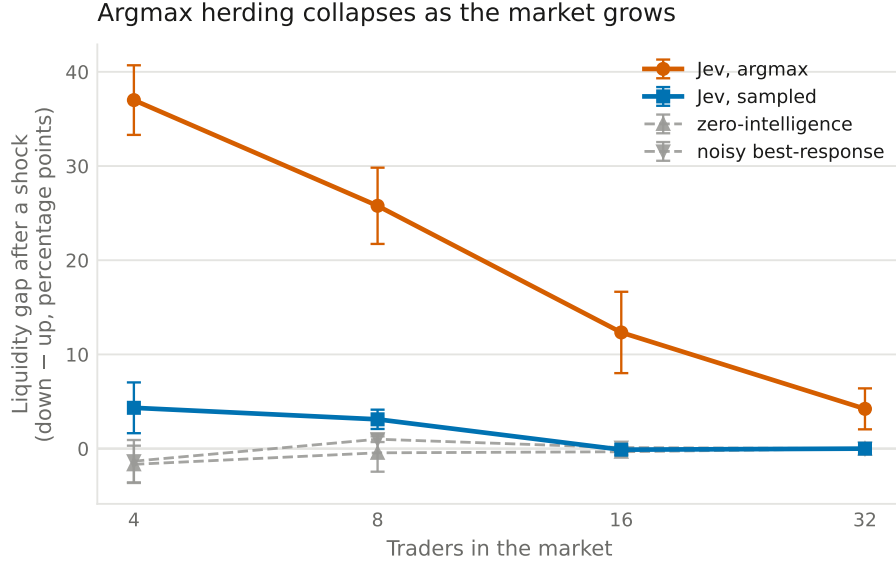


Figure 2: The liquidity gap against market size, twenty seeds per point, bars are standard errors. At $N = 32$ both symmetric baselines trade in 100% of periods with a zero gap, so there is no structural floor and the residual 4.2 points is attributable to the arm.

The failure is not a reasoning failure. The model’s modal action was correct on both sides at every mispricing tested. Nothing in its judgement was wrong; the loss occurred in aggregation. Evaluation of a single agent’s decisions cannot detect this, which is what makes it worth reporting.

9 Limitations

One unexplained asymmetry. The halt is markedly worse after positive shocks than negative ones, and we cannot explain it. Both symmetric baselines show no such lopsidedness at any market size, so it is not an artefact of the market’s construction. It is real, replicated, and unexplained.

One model. All results are for a single typed-decision model. We have no evidence about others, and cannot say whether the agreement rate that drives the mechanism is characteristic or idiosyncratic.

Provenance of the probabilities. We treat the returned probabilities as the model’s beliefs. We have asked the vendor whether they are calibrated posterior mass or post-processed, and have no answer. If the latter, Experiment 1 is a statement about a serving stack rather than a model.

Full information only. Every trader observes F_t exactly, which maximises the correlation the mechanism exploits. The delayed and noisy treatments are implemented but unreported, and would plausibly weaken the effect.

Scale. Markets of four to thirty-two traders over 160 periods. The extrapolation to larger markets is the fitted decay, not a measurement.

10 Reproducibility

Because the model is non-deterministic and has no seed, code alone does not reproduce these numbers. Every response is archived—110,805 of them—and the analysis can be replayed against the archive with live calls disabled, in which case a cache miss raises rather than silently contacting the API. Total spend was approximately \$3.30 across roughly 95,000 live calls.

The preregistrations, including the falsified hypotheses and the reasons the outcome measure changed between Experiments 1 and 2, are in the repository with their commit timestamps.

References

- D. K. Gode and S. Sunder. Allocative efficiency of markets with zero-intelligence traders: Market as a partial substitute for individual rationality. *Journal of Political Economy*, 101(1):119–137, 1993.
- V. L. Smith. An experimental study of competitive market behavior. *Journal of Political Economy*, 70(2):111–137, 1962.